This document is being provided for the exclusive use of YUI CHEUNG at THE UNIVERSITY OF HONG KONG

Deutsche Bank
Research

Rating
Buy

North America
United States

TMT
Software

Company
Microsoft

Reuters	Bloomberg	Exchange	Ticker
MSFT.OQ	MSFT US	NSM	MSFT

Date
29 January 2026

Forecast Change

Price at 28 Jan 2026 (USD)	481.63
Price target	575.00
52-week range	542.07 - 354.56

F2Q - Leaning into the long game

Valuation & Risks

Brad Zelnick
Research Analyst
+1-212-250-8563

Bhavin Shah, CFA
Research Analyst
+1-212-250-6775

Daniel Knauff
Research Associate
+1-415-262-2041

Yash Kejriwal
Research Associate
+1-212-250-6774

Microsoft reported another solid result for F2Q, but it ultimately fell short of more lofty market expectations, in particular for Azure growth. While most investors continue to hone in on this as the most important KPI, mgmt. again made it clear Azure is bearing the brunt of current supply constraints as they prioritize GPU allocation in support of AI usage in first-party apps and R&D to accelerate product innovation in pursuit of higher margin, recurring software opportunities up the stack. So while Azure growth still came in 1pt above guidance at +38% y/y @CC, this was shy of what we believe was a buyside bogey of >39%+ and what mgmt. indicated would have been >40% had they allocated all incremental GPU capacity in F1H26 to Azure. Importantly, we'd highlight Azure is still growing well above the market, guided to sustain 37-38% growth in F3Q and we believe can likely persist near here for several more quarters off a higher base.

From a margin and LTV perspective we believe the application layer has and is still generally perceived as highly strategic ground on which Microsoft has incumbent advantages. But we'd argue it is also the last area to see tangible benefits as AI monetization works its way up the stack, and is more than ever viewed as at risk of disruption and/or devaluation, which is weighing on sentiment. That story has yet to play out in our view, so while foregoing Azure revenue now for high-margin application and consumer SW opportunity ahead seems a sensible long-term strategy, it puts the onus on Microsoft to deliver before it will get credit. We continue to see more evidence of steadily growing traction, including new disclosure this quarter of 15m paid M365 Copilot seats (accelerating q/q and +160% y/y), paid GitHub Copilot subs now up to 4.7m (+75% y/y) and >100k medical providers using Dragon Copilot, but likely not yet tangible enough in financial results for investors to feel like they are seeing high impact from an ROIC perspective.

Bringing it back to a high level perspective, we think some of the debate and nuance around near-term bogeys, while important, is missing a story of exceptional share gains and execution. Amidst a heavy investment cycle and supply constraints, the company is poised to deliver a third straight year of mid-teens+ growth in operating income (incl. y/y margin expansion) and non-GAAP EPS. With continued strong demand signal from backlog (Commercial RPO +28% y/y ex-OAI) we remain confident in the durability of top- and bottom-line trajectories over the next several years. We maintain our Buy rating while adjusting our target price to $575 (from $630) to reflect our updated estimates and thoughts on the business and market. Our TP implies ~28x our updated CY27E non-GAAP EPS.

Key changes

TP	630.00 to 575.00	↓	-8.7%
Revenue (USDm)	327,026.9 to 328,168.9	↑	0.3%

Source: Deutsche Bank

Distributed on: 29/01/2026 08:03:28 GMT

Deutsche Bank Securities Inc.
IMPORTANT RESEARCH DISCLOSURES AND ANALYST CERTIFICATIONS LOCATED IN APPENDIX 1. Deutsche Bank does and seeks to do business with companies covered in its research reports. Thus, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.

This document is being provided for the exclusive use of YUI CHEUNG at THE UNIVERSITY OF HONG KONG

What We Liked / What We're watching on the Quarter

Revenue/Bookings

(+) F2Q revenue of $81.3bn was 1.5% above the midpoint of guidance, with upside concentrated in PBP & IC segments. On a constant currency basis, growth ticked down ~1.5pts sequentially to +15% y/y.

(+) Commercial Bookings rose +228% y/y @cc. This was primarily driven by the previously disclosed larger Azure commit from OpenAI, alongside benefit from the recent Anthropic commitment and "healthy growth" across core annuity sales motions on a lower expiry base when adjusted for OpenAI contracts in the prior year. Commercial RPO ended the quarter +110% y/y to $625bn; ~45% of the total balance is now attributed to commits from OpenAI, with the remaining core 55% growing +28% y/y. Total Commercial cRPO now stands at +39% y/y, up from +21% a year prior.

Business Lines

(=) Azure growth of +38% y/y @CC was 1pt above guidance (37%), but likely a touch shy of buyside expectations that we believe were closer to ~+39%. Upside to guidance was a result of efficiency gains across the fleet enabling reallocation of some capacity from other segments to monetizable Azure demand. We underscore that these results were once again limited by supply constraints as mgmt. noted the need to balance increasing Azure demand with growing needs across 1P services, increasing allocation to R&D teams, and replacement of end-of-life equipment.

(=) M365 Commercial Cloud growth of +14% y/y came in line with the high end of guidance +13-14% y/y @cc, with growth once again driven by a combination of seat growth (+6% y/y) and ARPU expansion. ARPU expansion also continues to be attributed to E5 momentum and then M365 Copilot. This quarter was also the first time mgmt. disclosed M365 copilot paid seats, which sit at 15m as of F2Q.

(+) Server Products grew +1% y/y @CC, much better compared to guidance for a LSD-MSD decline driven by demand for Microsoft's hybrid solutions, with benefit from the launch of SQL Server 2025 and higher transactional purchasing ahead of memory price increases.

(=) LinkedIn +10% y/y @CC accelerated 1pt from F1Q and was consistent with +10% y/y guidance, driven by strength in Marketing Solutions.

(-) Search and News Advertising revenue (ex-TAC) +9% y/y @CC missed guidance for low double digits growth (down sequentially as growth rates normalize) following some execution challenges. Growth continues to be driven by higher search volume and continued benefit from 3P partnerships.

(+) High margin Windows OEM + Devices revenue of +1% y/y and flattish @cc was much better than guidance of MSD decline attributable to strength in Windows OEM (+5% y/y) with continued benefit from Windows 10 end of support and still-elevated inventory levels from increased purchasing ahead of memory price increases, partially offset by a decline in Devices, as expected.

Margins/EPS/CapEx

This document is being provided for the exclusive use of YUI CHEUNG at THE UNIVERSITY OF HONG KONG

(+) Operating margin of 47.1% was +1.8pts above guidance (45.3%) and +1.6pt y/y. Gross margin of 68.0% exceeded guidance of 67.0%, though was down -0.7pts y/y on scaling AI investments that were only partially offset by efficiency gains, particularly in Azure and M365 Commercial Cloud. OpEx of $17.02bn came in 2% below guidance ($17.35bn), supporting strong operating income growth at +21% y/y and +19% y/y @cc.

(+) Non-GAAP EPS of $4.14 also came in above consensus of $3.94. This excludes impact from accounting gains on equity method investments in OpenAI associated with its recapitalization this quarter.

(=) Capex (inclusive of capital leases) rose by ~$2.6bn sequentially to ~$37.5bn in F2Q, in line with guidance to be sequentially higher. Roughly two-thirds was spent on short-lived assets, up from roughly 50/50 last quarter and consistent with mgmt.'s messaging around capex mix shift.

What We Liked / What We're watching on the Guide

(=) F3Q revenue guidance of $81.2bn at the midpoint came in a touch below Street ($81.4bn); PBP and IC segment revenue guidance was ahead of consensus while MPC is seen declining y/y. FX is anticipated to be a 3pt tailwind to revenue in the quarter.

(+) F3Q Azure guidance of +37-38% y/y @CC was 1.5-2.5pt ahead of consensus ~35.5%, signaling stability despite a 4pt harder comp. and expectations for demand to remain significantly ahead of available capacity. Mgmt. noted that they will continue to balance servicing Azure demand with growing needs across 1P apps and AI solutions, MSFT R&D efforts, and end-of-life server replacements.

(=) F3Q M365 Commercial Cloud guidance for +13-14% y/y @CC is roughly in line with our prior expectation at ~13.5%. ARPU growth is again expected to be driven by E5 and M365 Copilot.

(=) Guidance for F3Q OM to tick down slightly by 0.6pts to ~45.1% was roughly in line with Street and assumes gross margins of 67.1% are down 1.7pts y/y and OpEx is +10.5% y/y.

(+) On a full-year basis, mgmt. now expects FY26 operating margins to be up "slightly" (vs. flat prior) given a strong 1H and benefit from higher mix of Windows OEM and Commercial on-prem revenue. Additionally, from an FX perspective, 4Q total revenue and COGS are expected to increase <1pt, with roughly a ~1pt revenue benefit in PBP and MPC, and <1pt in IC.

(=) F3Q Capex is expected to be sequentially lower following normal variability from cloud infrastructure buildouts and timing delivery of finance leases, with mix of short-lived assets expected to be similar to F2Q. Separately, mgmt. noted rising memory prices could impact capital expenditures for the full year; however impact to Microsoft Cloud gross margins will be phased given the 6-year depreciation life of these assets.

Estimate Revisions and Valuation

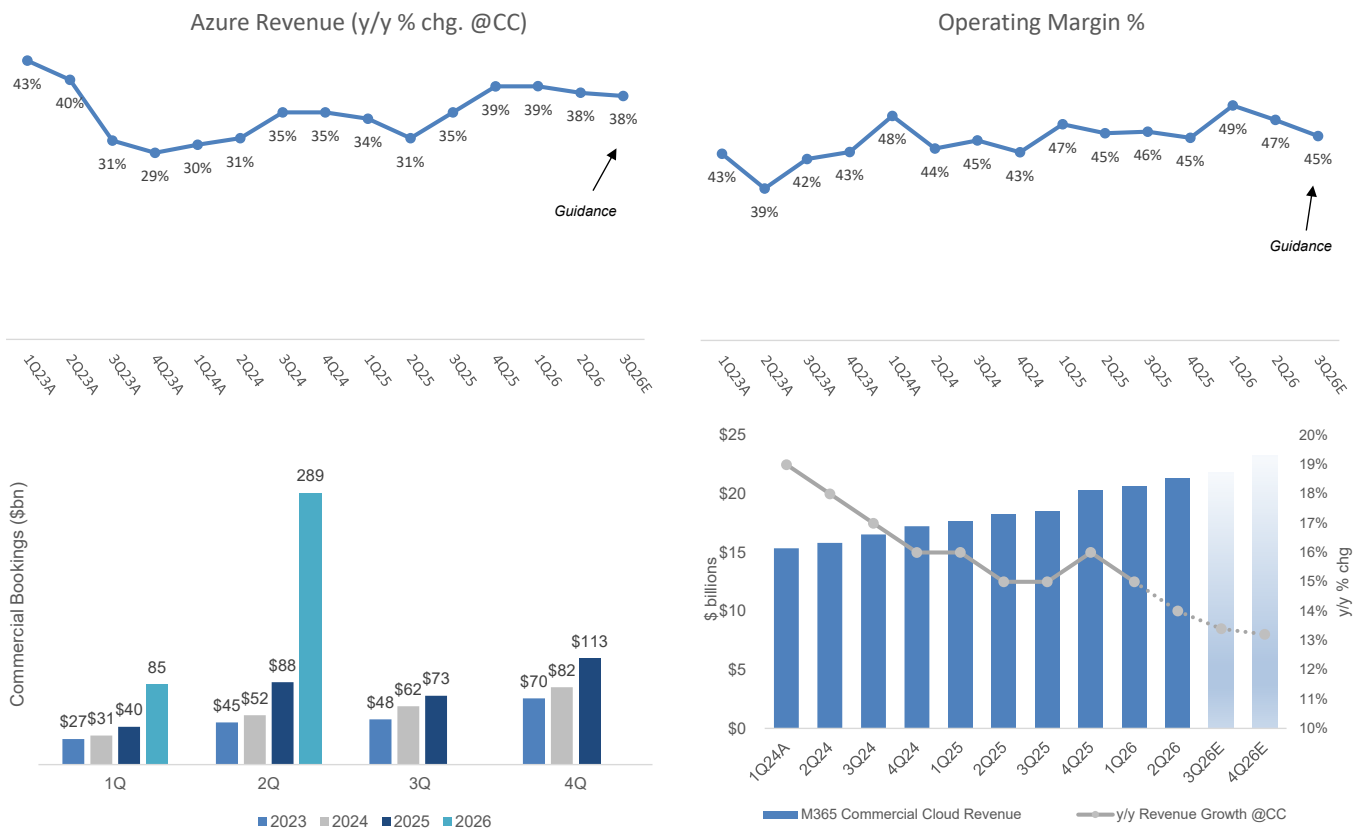
Incorporating F2Q results, guidance and our latest thoughts on the business, we increase FY26/27 revenue estimates to $328.2/$377.6bn from $327.0/$375.2bn and non-GAAP EPS estimates to $16.60/$19.10 from $16.43/$19.01, respectively.

This document is being provided for the exclusive use of YUI CHEUNG at THE UNIVERSITY OF HONG KONG

We reduce our TP to $575 (from $630), implying a P/E of ~28x our CY27E EPS. We use a DCF to arrive at our target price. Our DCF is based on a WACC of 9.1%, a risk-free rate of 4.0% and an equity risk premium of 5.3%. We use a terminal growth rate of 3.5%, based on GDP growth. Downside risks to our Buy rating include: (1) Increased competition for AI and public cloud; (2) Further deterioration of the macro environment; and (3) Expectations for continued growth at large scale.

This document is being provided for the exclusive use of YUI CHEUNG at THE UNIVERSITY OF HONG KONG.

Figure 1:Key Charts



Source : Company Reports, DB Estimates

This document is being provided for the exclusive use of YUI CHEUNG at THE UNIVERSITY OF HONG KONG

Figure 2: F2Q26 Summary

Microsoft (MSFT)
In $mn except EPS

	2Q26				
	Actual	DBe	Actual vs. DBe	Cons.	Company guidance
Income Statement					
Productivity & Business Process	34,116	33,496	+1.9%	33,481	33,300 - 33,600
Y/Y Growth	*+15.9%*	*+13.8%*		*+13.7%*	
Q/Q Growth	*+3.3%*	*+1.4%*		*+1.4%*	
Intelligent Cloud	32,907	32,399	+1.6%	32,420	32,250 - 32,550
Y/Y Growth	*+28.8%*	*+26.8%*		*+26.9%*	
Q/Q Growth	*+6.5%*	*+4.9%*		*+4.9%*	
More Personal Computing	14,250	14,219	+0.2%	14,279	13,950 - 14,450
Y/Y Growth	*-2.7%*	*-2.9%*		*-2.5%*	
Q/Q Growth	*+3.6%*	*+3.4%*		*+3.8%*	
Total Revenue	**81,273**	**80,114**	**+1.4%**	**80,289**	**79,500 - 80,600**
Y/Y Growth	*+16.7%*	*+15.1%*		*+15.3%*	
Q/Q Growth	*+4.6%*	*+3.1%*		*+3.4%*	
Gross Profit	55,295	53,644	+3.1%	53,711	52,950 - 54,250
GM (%)	68.0%	67.0%		66.9%	
Operating Income	**38,275**	**36,292**	**+5.5%**	**36,495**	**35,550 - 36,950**
OM (%)	47.1%	45.3%		45.5%	
Y/Y Growth	*+20.9%*	*+14.7%*		*+15.3%*	
Non-GAAP Net Income	30,875	29,477	+4.7%	29,406	
Net margin (%)	38.0%	36.8%		36.6%	
Non-GAAP EPS	**$4.14**	**$3.95**	**+4.7%**	**$3.94**	
Operating Metrics					
Operating Cash Flow	**35,758**	**24,565**	**+45.6%**	**28,630**	
Y/Y Growth	*+60.4%*	*+10.2%*		*+28.4%*	
Levered Free Cash Flow	5,882	531	+1007.5%		

Source : DB estimates, Company data, Bloomberg Finance LP

This document is being provided for the exclusive use of YUI CHEUNG at THE UNIVERSITY OF HONG KONG.

Figure 3: Estimates Summary

Microsoft (MSFT)

In $mn except EPS

	3Q26E					FY26E				FY27E			
	New Estimates	Old Estimates	New vs. Old	Cons.	Company guidance	New Estimates	Old Estimates	New vs. Old	Cons.	New Estimates	Old Estimates	New vs. Old	Cons.
Income Statement													
Productivity & Business Process	34,309	33,967	+1.0%	33,953	34,250 - 34,550	138,415	137,071	+1.0%	137,031	152,649	150,760	+1.3%	150,600
Y/Y Growth	*+14.6%*	*+13.4%*		*+13.4%*		*+14.6%*	*+13.5%*		*+13.4%*	*+10.3%*	*+10.0%*		*+9.9%*
Intelligent Cloud	34,250	34,099	+0.4%	33,760	34,100 - 34,400	136,139	134,795	+1.0%	133,887	170,473	167,754	+1.6%	170,796
Y/Y Growth	*+28.0%*	*+27.5%*		*+26.2%*		*+28.1%*	*+26.8%*		*+26.0%*	*+25.2%*	*+24.5%*		*+27.6%*
More Personal Computing	12,629	13,522	-6.6%	13,612	12,300 - 12,800	53,615	55,161	-2.8%	55,420	54,435	56,691	-4.0%	57,080
Y/Y Growth	*-5.5%*	*+1.1%*		*+1.8%*		*-1.9%*	*+0.9%*		*+1.4%*	*+1.5%*	*+2.8%*		*+3.0%*
Total Revenue	**81,189**	**81,588**	**-0.5%**	**81,360**	**80,650 - 81,750**	**328,169**	**327,027**	**+0.3%**	**326,500**	**377,556**	**375,205**	**+0.6%**	**375,692**
Y/Y Growth	*+15.9%*	*+16.4%*		*+16.1%*		*+16.5%*	*+16.1%*		*+15.9%*	*+15.0%*	*+14.7%*		*+15.1%*
Gross Profit	54,478	54,664	-0.3%	54,734	53,800 - 55,100	222,474	220,954	+0.7%	221,334	251,794	250,230	+0.6%	252,587
GM (%)	67.1%	67.0%		67.3%		67.8%	67.6%		67.8%	66.7%	66.7%		67.2%
Operating Income	**36,632**	**37,245**	**-1.6%**	**37,007**	**35,900 - 37,300**	**152,088**	**151,011**	**+0.7%**	**150,593**	**174,778**	**173,926**	**+0.5%**	**173,875**
OM (%)	45.1%	45.7%		45.5%		46.3%	46.2%		46.1%	46.3%	46.4%		46.3%
Y/Y Growth	*+14.5%*	*+16.4%*		*+15.6%*		*+18.3%*	*+17.5%*		*+17.2%*	*+14.9%*	*+15.2%*		*+15.5%*
Non-GAAP Net Income	30,239	30,168	+0.2%	29,529		123,715	122,485	+1.0%	119,628	141,570	140,880	+0.5%	140,235
Net margin (%)	37.2%	37.0%		36.3%		37.7%	37.5%		36.6%	37.5%	37.5%		37.3%
Non-GAAP EPS	**$4.06**	**$4.05**	**+0.2%**	**$3.99**		**$16.60**	**$16.43**	**+1.0%**	**$15.99**	**$19.10**	**$19.01**	**+0.5%**	**$18.78**
Operating Metrics													
Operating Cash Flow	**42,877**	**40,302**	**+6.4%**	**42,501**		**172,346**	**161,895**	**+6.5%**	**165,656**	**200,062**	**197,820**	**+1.1%**	**197,167**
Y/Y Growth	*+15.7%*	*+8.8%*		*+14.7%*		*+26.6%*	*+18.9%*		*21.7%*	*+16.1%*	*+22.2%*		*+19.0%*
Levered Free Cash Flow	18,520	15,826	+17.0%			72,310	67,695	+6.8%		87,769	86,228	+1.8%	

Source : DB estimates, Company data, Bloomberg Finance LP

This document is being provided for the exclusive use of YUI CHEUNG at THE UNIVERSITY OF HONG KONG

Appendix 1

Important Disclosures

*Other information available upon request

Disclosure checklist

Company	Ticker	Recent price*	Disclosure
Microsoft	MSFT.OQ	481.63 (USD) 28 Jan 2026	7, 8, 14, 15, 21, 24, 26

*Prices are current as of the end of the previous trading session unless otherwise indicated and are sourced from local exchanges via Reuters, Bloomberg and other vendors . Other information is sourced from Deutsche Bank, subject companies, and other sources. For disclosures pertaining to recommendations or estimates made on securities other than the primary subject of this research, please see the most recently published company report or visit our global disclosure look-up page on our website at https://research.db.com/Research/Disclosures/EquityResearchDisclosures. Aside from within this report, important risk and conflict disclosures can also be found at https://research.db.com/Research/Disclosures/Disclaimer. Investors are strongly encouraged to review this information before investing.

Important Disclosures Required by U.S. Regulators

Disclosures marked with an asterisk may also be required by at least one jurisdiction in addition to the United States.See Important Disclosures Required by Non-US Regulators and Explanatory Notes.

7. Deutsche Bank and/or its affiliate(s) has received compensation from this company for the provision of investment banking or financial advisory services within the past year.
8. Deutsche Bank and/or its affiliate(s) expects to receive, or intends to seek, compensation for investment banking services from this company in the next three months.
14. Deutsche Bank and/or its affiliate(s) has received compensation from this company within the past year for non-investment banking related services.
15. This company has been a client of Deutsche Bank Securities Inc. within the past year during which time it received investment banking services.
21. This company has been a client of Deutsche Bank Securities Inc. within the past year, during which time it received non-investment banking securities-related services.

Important Disclosures Required by Non-U.S. Regulators

Disclosures marked with an asterisk may also be required by at least one jurisdiction in addition to the United States.See Important Disclosures Required by Non-US Regulators and Explanatory Notes.

24. Deutsche Bank and/or its affiliate(s) is or has been over the previous 12 months party to an agreement with the company relating to the provision of services set out in Sections A and B of Annex I of Directive 2014/65/EU, or has over the previous 12 months been obliged or entitled (as applicable) to pay or receive compensation relating to the provision of services set out in Sections A and B of Annex I of Directive 2014/65/EU.
26. Within the preceding 12 months, Deutsche Bank and/or its affiliate(s) has received compensation for the provision of investment banking services or is currently providing or has provided investment banking services to this company.

For disclosures pertaining to recommendations or estimates made on securities other than the primary subject of this research, please see the most recently published company report or visit our global disclosure look-up page on our website at https://research.db.com/Research/Disclosures/EquityResearchDisclosures. Aside from within this report, important risk and conflict disclosures can also be found at https://research.db.com/Research/Disclosures/Disclaimer. Investors are strongly encouraged to review this information before investing.

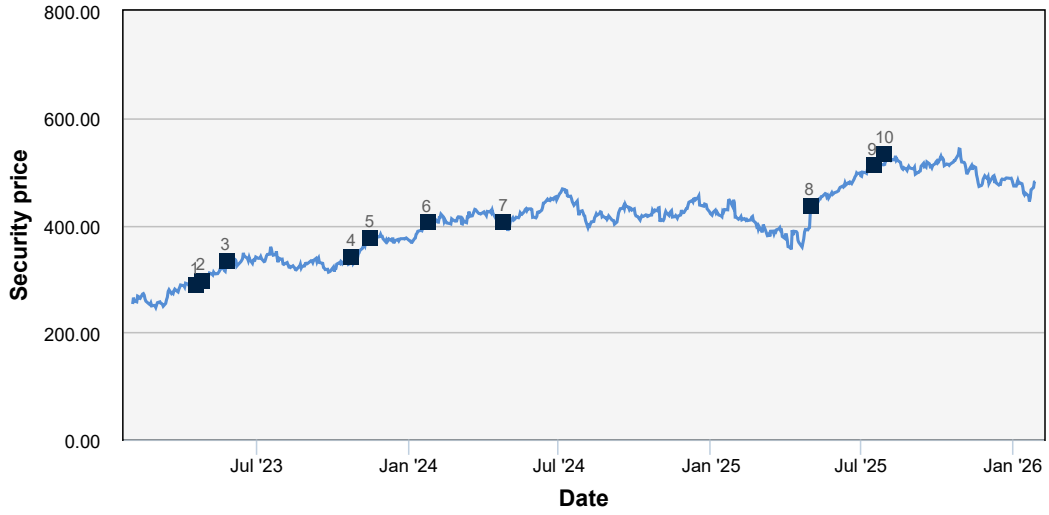
Analyst Certification

The views expressed in this report accurately reflect the personal views of the undersigned lead analyst(s) about the subject issuer and the securities of the issuer. In addition, the undersigned lead analyst(s) has not and will not receive any compensation for providing a specific recommendation or view in this report. Brad Zelnick.

This document is being provided for the exclusive use of YUI CHEUNG at THE UNIVERSITY OF HONG KONG

Historical recommendations and target price: Microsoft (MSFT.OQ)

(as of 01/28/2026)



Current Recommendations

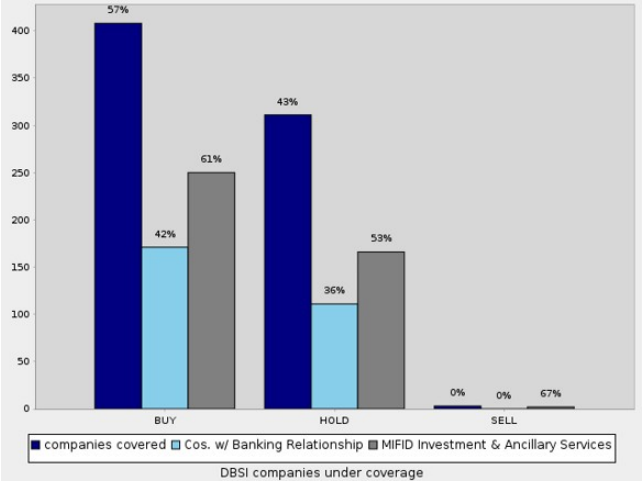
- Buy
- Hold
- Sell
- Not Rated
- Suspended Rating

** Analyst is no longer at Deutsche Bank

1.	04/19/2023	Buy, Target Price Change USD 310.00, Current Price USD 288.45 Brad Zelnick
2.	04/26/2023	Buy, Target Price Change USD 340.00, Current Price USD 295.37 Brad Zelnick
3.	05/26/2023	Buy, Target Price Change USD 380.00, Current Price USD 332.89 Brad Zelnick
4.	10/25/2023	Buy, Target Price Change USD 395.00, Current Price USD 340.67 Brad Zelnick
5.	11/16/2023	Buy, Target Price Change USD 415.00, Current Price USD 376.17 Brad Zelnick
6.	01/25/2024	Buy, Target Price Change USD 450.00, Current Price USD 404.87 Brad Zelnick
7.	04/26/2024	Buy, Target Price Change USD 475.00, Current Price USD 406.32 Brad Zelnick
8.	05/02/2025	Buy, Target Price Change USD 500.00, Current Price USD 435.28 Brad Zelnick
9.	07/17/2025	Buy, Target Price Change USD 550.00, Current Price USD 511.70 Brad Zelnick
10.	07/31/2025	Buy, Target Price Change USD 630.00, Current Price USD 533.50 Brad Zelnick

This document is being provided for the exclusive use of YUI CHEUNG at THE UNIVERSITY OF HONG KONG

Equity rating dispersion and banking relationships



Equity Rating and Dispersion Key

The Equity Rating Dispersion Chart depicts the following:

The proportion of recommendations that are rated "buy", "sell" and "hold" over the previous 12 months. This is shown for securities issued in the stated region e.g. "Europe Universe". See rating definitions below. This is represented by the "Companies Covered" bars in the chart. The percentage value displayed above the bar is the proportion as a percentage. E.g. 50% above the “buy” / “Companies Covered” bar means that 50% of DB’s equity research covered companies over the past 12 months have a “buy” rating.

Next to each of the three respective bars showing the proportion of "buy", "sell" and "hold" recommendations we provide two additional bars to show:

- The proportion of "buy", "sell" or "hold recommendations where Deutsche Bank and or/Affiliates provided MIFID Investment or Ancillary Services in the past 12 months. This is represented in the "MIFID Investment and Ancillary Services" bar. The percentage value displayed above the bar shows the proportion of Companies Covered with the given rating where DB has also provided MIFID Investment and Ancillary Services in the past 12 months. E.g. 50% above the “Cos. w/ MIFID Investment and Ancillary Services” bar means 50% of the Companies Covered with the rating stated have also received MIFID Investment and Ancillary Services from DB.

- The proportion of "buy" (or "sell" or "hold) recommendations where Deutsche Bank and or/Affiliates has provided Investment Banking services in the past 12 months for which it has received compensation. The percentage value displayed above the bar shows the proportion of Companies Covered with the stated rating where DB has also provided Investment Banking services in the past 12 months. E.g. 50% above the “Cos. w/ Investment Banking relationship” bar means 50% of the Companies Covered with the rating stated also have an Investment Banking Relationship with DB.

Buy: Based on a current 12- month view of TSR, we recommend that investors buy the stock.

Sell: Based on a current 12-month view of TSR, we recommend that investors sell the stock.

Hold: We take a neutral view on the stock 12-months out and, based on this time horizon, do not recommend either a Buy or Sell.

TSR = Total Shareholder Return. Percentage change in share price from current price to projected target price plus projected dividend yield

Newly issued research recommendations and target prices supersede previously published research.

This document is being provided for the exclusive use of YUI CHEUNG at THE UNIVERSITY OF HONG KONG

Additional Information

The information and opinions in this report were prepared by Deutsche Bank AG or one of its affiliates (collectively 'Deutsche Bank'). Though the information herein is believed to be reliable and has been obtained from public sources believed to be reliable, Deutsche Bank makes no representation as to its accuracy or completeness. Hyperlinks to third-party websites in this report are provided for reader convenience only. Deutsche Bank neither endorses the content nor is responsible for the accuracy or security controls of those websites.

If you use the services of Deutsche Bank in connection with a purchase or sale of a security that is discussed in this report, or is included or discussed in another communication (oral or written) from a Deutsche Bank analyst, Deutsche Bank may act as principal for its own account or as agent for another person.

Deutsche Bank may consider this report in deciding to trade as principal. It may also engage in transactions, for its own account or with customers, in a manner inconsistent with the views taken in this research report. Others within Deutsche Bank, including strategists, sales staff and other analysts, may take views that are inconsistent with those taken in this research report. Deutsche Bank issues a variety of research products, including fundamental analysis, equity-linked analysis, quantitative analysis and trade ideas. Recommendations contained in one type of communication may differ from recommendations contained in others, whether as a result of differing time horizons, methodologies, perspectives or otherwise. Deutsche Bank and/or its affiliates may also be holding debt or equity securities of the issuers it writes on. Analysts are paid in part based on the profitability of Deutsche Bank AG and its affiliates, which includes investment banking, trading and principal trading revenues.

Opinions, estimates and projections constitute the current judgment of the author as of the date of this report. They do not necessarily reflect the opinions of Deutsche Bank and are subject to change without notice. Deutsche Bank provides liquidity for buyers and sellers of securities issued by the companies it covers. Deutsche Bank research analysts sometimes have shorter-term trade ideas that may be inconsistent with Deutsche Bank's existing longer-term ratings. Some trade ideas for equities are listed as Catalyst Calls on the Research Website (https://research.db.com/Research/) , and can be found on the general coverage list and also on the covered company's page. A Catalyst Call represents a high-conviction belief by an analyst that a stock will outperform or underperform the market and/or a specified sector over a time frame of no less than two weeks and no more than three months. In addition to Catalyst Calls, analysts may occasionally discuss with our clients, and with Deutsche Bank salespersons and traders, trading strategies or ideas that reference catalysts or events that may have a near-term or medium-term impact on the market price of the securities discussed in this report, which impact may be directionally counter to the analysts' current 12-month view of total return or investment return as described herein. Deutsche Bank has no obligation to update, modify or amend this report or to otherwise notify a recipient thereof if an opinion, forecast or estimate changes or becomes inaccurate. Coverage and the frequency of changes in market conditions and in both general and company-specific economic prospects make it difficult to update research at defined intervals. Updates are at the sole discretion of the coverage analyst or of the Research Department Management, and the majority of reports are published at irregular intervals. This report is provided for informational purposes only and does not take into account the particular investment objectives, financial situations, or needs of individual clients. It is not an offer or a solicitation of an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Target prices are inherently imprecise and a product of the analyst's judgment. The financial instruments discussed in this report may not be suitable for all investors, and investors must make their own informed investment decisions. Prices and availability of financial instruments are subject to change without notice, and investment transactions can lead to losses as a result of price fluctuations and other factors. If a financial instrument is denominated in a currency other than an investor's currency, a change in exchange rates may adversely affect the investment. Past performance is not necessarily indicative of future results. Performance calculations exclude transaction costs, unless otherwise indicated. Unless otherwise indicated, prices are current as of the end of the previous trading session and are sourced from local exchanges via Reuters, Bloomberg and other vendors. Data is also sourced from Deutsche Bank, subject companies, and other parties. Artificial intelligence tools may be used in the preparation of this material, including but not limited to assist in fact-finding, data analysis, pattern recognition, content drafting and editorial corrections pertaining to research material.

The Deutsche Bank Research Department is independent of other business divisions of the Bank. Details regarding our organizational arrangements and information barriers we have to prevent and avoid conflicts of interest with respect to our research are available on our website (https://research.db.com/Research/) under Disclaimer.

Macroeconomic fluctuations often account for most of the risks associated with exposures to instruments that promise to pay fixed or variable interest rates. For an investor who is long fixed-rate instruments (thus receiving these cash flows), increases in interest rates naturally lift the discount factors applied to the expected cash flows and thus cause a loss. The longer the maturity of a certain cash flow and the higher the move in the discount factor, the higher will be the loss. Upside surprises in inflation, fiscal funding needs, and FX depreciation rates are among the most common adverse macroeconomic shocks to receivers. But counterparty exposure, issuer creditworthiness, client segmentation, regulation (including changes in assets holding limits for different types of investors), changes in tax policies, currency convertibility (which may constrain currency conversion, repatriation of profits and/or liquidation of positions), and settlement issues related to local clearing houses are also important risk factors. The sensitivity of fixed-income instruments to macroeconomic shocks may be mitigated by indexing the contracted cash flows to inflation, to FX depreciation, or to specified interest rates - these are common in emerging markets. The index fixings may - by construction - lag or mis-measure the actual move in the underlying variables they are intended to track. The choice of the proper fixing (or metric) is particularly important in swaps markets, where floating coupon rates (i.e., coupons indexed to a typically short-dated interest rate reference index) are exchanged for fixed coupons. Funding in a currency that differs from the currency in which coupons are denominated carries FX risk. Options on swaps (swaptions) the risks typical to options in addition to the risks related to rates movements.

This document is being provided for the exclusive use of YUI CHEUNG at THE UNIVERSITY OF HONG KONG

Derivative transactions involve numerous risks including market, counterparty default and illiquidity risk. The appropriateness of these products for use by investors depends on the investors' own circumstances, including their tax position, their regulatory environment and the nature of their other assets and liabilities; as such, investors should take expert legal and financial advice before entering into any transaction similar to or inspired by the contents of this publication. The risk of loss in futures trading and options, foreign or domestic, can be substantial. As a result of the high degree of leverage obtainable in futures and options trading, losses may be incurred that are greater than the amount of funds initially deposited - up to theoretically unlimited losses. Trading in options involves risk and is not suitable for all investors. Prior to buying or selling an option, investors must review the 'Characteristics and Risks of Standardized Options", at https://www.theocc.com/company-information/documents-and-archives/options-disclosure-document. If you are unable to access the website, please contact your Deutsche Bank representative for a copy of this important document.

Participants in foreign exchange transactions may incur risks arising from several factors, including the following: (i) exchange rates can be volatile and are subject to large fluctuations; (ii) the value of currencies may be affected by numerous market factors, including world and national economic, political and regulatory events, events in equity and debt markets and changes in interest rates; and (iii) currencies may be subject to devaluation or government-imposed exchange controls, which could affect the value of the currency. Investors in securities such as ADRs, whose values are affected by the currency of an underlying security, effectively assume currency risk.

Unless governing law provides otherwise, all transactions should be executed through the Deutsche Bank entity in the investor's home jurisdiction. Aside from within this report, important conflict disclosures can also be found at https://research.db.com/Research/ on each company's research page or under the 'Disclosures' tab. Investors are strongly encouraged to review this information before investing.

Deutsche Bank (which includes Deutsche Bank AG, its branches and affiliated companies) is not acting as a financial adviser, consultant or fiduciary to you or any of your agents (collectively, "You" or "Your") with respect to any information provided in this report. Deutsche Bank does not provide investment, legal, tax or accounting advice, Deutsche Bank is not acting as your impartial adviser, and does not express any opinion or recommendation whatsoever as to any strategies, products or any other information presented in the materials. Information contained herein is being provided solely on the basis that the recipient will make an independent assessment of the merits of any investment decision, and it does not constitute a recommendation of, or express an opinion on, any product or service or any trading strategy.

The information presented is general in nature and is not directed to retirement accounts or any specific person or account type, and is therefore provided to You on the express basis that it is not advice, and You may not rely upon it in making Your decision. The information we provide is being directed only to persons we believe to be financially sophisticated, who are capable of evaluating investment risks independently, both in general and with regard to particular transactions and investment strategies, and who understand that Deutsche Bank has financial interests in the offering of its products and services. If this is not the case, or if You are an IRA or other retail investor receiving this directly from us, we ask that you inform us immediately.

In July 2018, Deutsche Bank revised its rating system for short term ideas whereby the branding has been changed to Catalyst Calls ("CC") from SOLAR ideas; the rating categories for Catalyst Calls originated in the Americas region have been made consistent with the categories used by Analysts globally; and the effective time period for CCs has been reduced from a maximum of 180 days to 90 days.

United States: Approved and/or distributed by Deutsche Bank Securities Incorporated, a member of FINRA and SIPC. Analysts located outside of the United States are employed by non-US affiliates and are not registered/qualified as research analysts with FINRA.

European Economic Area (exc. United Kingdom): Approved and/or distributed by Deutsche Bank AG, a joint stock corporation with limited liability incorporated in the Federal Republic of Germany with its principal office in Frankfurt am Main. Deutsche Bank AG is authorized under German Banking Law and is subject to supervision by the European Central Bank and by BaFin, Germany's Federal Financial Supervisory Authority.

United Kingdom: Approved and/or distributed by Deutsche Bank AG acting through its London Branch at 21 Moorfields, London EC2Y 9DB. Deutsche Bank AG in the United Kingdom is authorised by the Prudential Regulation Authority and is subject to limited regulation by the Prudential Regulation Authority and Financial Conduct Authority. Details about the extent of our authorisation and regulation are available on request.

Hong Kong SAR: Distributed by Deutsche Bank AG, Hong Kong Branch except for any research content relating to futures contracts within the meaning of the Hong Kong Securities and Futures Ordinance Cap. 571. Research reports on such futures contracts are not intended for access by persons who are located, incorporated, constituted or resident in Hong Kong. The author(s) of a research report may not be licensed to carry on regulated activities in Hong Kong and, if not licensed, do not hold themselves out as being able to do so. The provisions set out above in the 'Additional Information' section shall apply to the fullest extent permissible by local laws and regulations, including without limitation the Code of Conduct for Persons Licensed or Registered with the Securities and Futures Commission. This report is intended for distribution only to 'professional investors' as defined in Part 1 of Schedule of the SFO. This document must not be acted or relied on by persons who are not professional investors. Any investment or investment activity to which this document relates is only available to professional investors and will be engaged only with professional investors.

India: Prepared by Deutsche Equities India Private Limited (DEIPL) having CIN: U65990MH2002PTC137431 and registered

This document is being provided for the exclusive use of YUI CHEUNG at THE UNIVERSITY OF HONG KONG

office at 14th Floor, The Capital, C-70, G Block, Bandra Kurla Complex, Mumbai (India) 400051. Tel: + 91 22 7180 4444. It is registered by the Securities and Exchange Board of India (SEBI) as a Stock broker bearing registration no.: INZ000252437; Merchant Banker bearing SEBI Registration no.: INM000010833 and Research Analyst bearing SEBI Registration no.: INH000001741. DEIPL's Compliance / Grievance officer is Ms. Rashmi Poddar (Tel: +91 22 7180 4929 email ID: complaints.deipl@db.com). Registration granted by SEBI and certification from NISM in no way guarantee performance of DEIPL or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. DEIPL may have received administrative warnings from the SEBI for breaches of Indian regulations. Deutsche Bank and/or its affiliate(s) may have debt holdings or positions in the subject company. With regard to information on associates, please refer to the "Shareholdings" section in the Annual Report at: https://www.db.com/ir/en/annual-reports.htm. For latest India research audit report, refer https://country.db.com/india/deutsche-equities-india/index?language_id=1.

Japan: Approved and/or distributed by Deutsche Securities Inc.(DSI). Registration number - Registered as a financial instruments dealer by the Head of the Kanto Local Finance Bureau (Kinsho) No. 117. Member of associations: JSDA, Type II Financial Instruments Firms Association and The Financial Futures Association of Japan. Commissions and risks involved in stock transactions - for stock transactions, we charge stock commissions and consumption tax by multiplying the transaction amount by the commission rate agreed with each customer. Stock transactions can lead to losses as a result of share price fluctuations and other factors. Transactions in foreign stocks can lead to additional losses stemming from foreign exchange fluctuations. We may also charge commissions and fees for certain categories of investment advice, products and services. Recommended investment strategies, products and services carry the risk of losses to principal and other losses as a result of changes in market and/or economic trends, and/or fluctuations in market value. Before deciding on the purchase of financial products and/or services, customers should carefully read the relevant disclosures, prospectuses and other documentation. 'Moody's', 'Standard Poor's', and 'Fitch' mentioned in this report are not registered credit rating agencies in Japan unless Japan or 'Nippon' is specifically designated in the name of the entity. Reports on Japanese listed companies not written by analysts of DSI are written by Deutsche Bank Group's analysts with the coverage companies specified by DSI. Some of the foreign securities stated on this report are not disclosed according to the Financial Instruments and Exchange Law of Japan. Target prices set by Deutsche Bank's equity analysts are based on a 12-month forecast period.

Korea: Distributed by Deutsche Securities Korea Co.

South Africa: Deutsche Bank AG Johannesburg is incorporated in the Federal Republic of Germany (Branch Register Number in South Africa: 1998/003298/10).

Singapore: This report is issued by Deutsche Bank AG, Singapore Branch (One Raffles Quay #18-00 South Tower Singapore 048583, 65 6423 8001), which may be contacted in respect of any matters arising from, or in connection with, this report. Where this report is issued or promulgated by Deutsche Bank in Singapore to a person who is not an accredited investor, expert investor or institutional investor (as defined in the applicable Singapore laws and regulations), they accept legal responsibility to such person for its contents.

Taiwan: Information on securities/investments that trade in Taiwan is for your reference only. Readers should independently evaluate investment risks and are solely responsible for their investment decisions. Deutsche Bank research may not be distributed to the Taiwan public media or quoted or used by the Taiwan public media without written consent. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation to trade in such securities/instruments.

Qatar: Deutsche Bank AG in the Qatar Financial Centre (registered no. 00032) is regulated by the Qatar Financial Centre Regulatory Authority. Deutsche Bank AG - QFC Branch may undertake only the financial services activities that fall within the scope of its existing QFCRA license. Its principal place of business in the QFC: Qatar Financial Centre, Tower, West Bay, Level 5, PO Box 14928, Doha, Qatar. This information has been distributed by Deutsche Bank AG. Related financial products or services are only available only to Business Customers, as defined by the Qatar Financial Centre Regulatory Authority.

Russia: The information, interpretation and opinions submitted herein are not in the context of, and do not constitute, any appraisal or evaluation activity requiring a license in the Russian Federation.

Kingdom of Saudi Arabia: Deutsche Securities Saudi Arabia (DSSA) is a closed joint stock company authorized by the Capital Market Authority of the Kingdom of Saudi Arabia with a license number (No. 37-07073) to conduct the following business activities: Dealing, Arranging, Advising, and Custody activities. DSSA registered office is Faisaliah Tower, 17th Floor, King Fahad Road - Al Olaya District Riyadh, Kingdom of Saudi Arabia P.O. Box 301806.

United Arab Emirates: Deutsche Bank AG in the Dubai International Financial Centre (registered no. 00045) is regulated by the Dubai Financial Services Authority. Deutsche Bank AG - DIFC Branch may only undertake the financial services activities that fall within the scope of its existing DFSA license. Principal place of business in the DIFC: Dubai International Financial Centre, The Gate Village, Building 5, PO Box 504902, Dubai, U.A.E. This information has been distributed by Deutsche Bank AG. Related financial products or services are available only to Professional Clients, as defined by the Dubai Financial Services Authority.

Australia and New Zealand: This research is intended only for 'wholesale clients' within the meaning of the Australian Corporations Act and New Zealand Financial Advisors Act, respectively. Please refer to Australia-specific research disclosures and related information at https://www.dbresearch.com/PROD/RPS_EN-PROD/PROD0000000000521304.xhtml . Where

This document is being provided for the exclusive use of YUI CHEUNG at THE UNIVERSITY OF HONG KONG

research refers to any particular financial product recipients of the research should consider any product disclosure statement, prospectus or other applicable disclosure document before making any decision about whether to acquire the product. In preparing this report, the primary analyst or an individual who assisted in the preparation of this report has likely been in contact with the company that is the subject of this research for confirmation/clarification of data, facts, statements, permission to use company-sourced material in the report, and/or site-visit attendance. Without prior approval from Research Management, analysts may not accept from current or potential Banking clients the costs of travel, accommodations, or other expenses incurred by analysts attending site visits, conferences, social events, and the like. Similarly, without prior approval from Research Management and Anti-Bribery and Corruption ("ABC") team, analysts may not accept perks or other items of value for their personal use from issuers they cover.

Additional information relative to securities, other financial products or issuers discussed in this report is available upon request. This report may not be reproduced, distributed or published without Deutsche Bank's prior written consent.

Backtested, hypothetical or simulated performance results have inherent limitations. Unlike an actual performance record based on trading actual client portfolios, simulated results are achieved by means of the retroactive application of a backtested model itself designed with the benefit of hindsight. Taking into account historical events the backtesting of performance also differs from actual account performance because an actual investment strategy may be adjusted any time, for any reason, including a response to material, economic or market factors. The backtested performance includes hypothetical results that do not reflect the reinvestment of dividends and other earnings or the deduction of advisory fees, brokerage or other commissions, and any other expenses that a client would have paid or actually paid. No representation is made that any trading strategy or account will or is likely to achieve profits or losses similar to those shown. Alternative modeling techniques or assumptions might produce significantly different results and prove to be more appropriate. Past hypothetical backtest results are neither an indicator nor guarantee of future returns. Actual results will vary, perhaps materially, from the analysis.

The method for computing individual E,S,G and composite ESG scores set forth herein is a novel method developed by the Research department within Deutsche Bank AG, computed using a systematic approach without human intervention. Different data providers, market sectors and geographies approach ESG analysis and incorporate the findings in a variety of ways. As such, the ESG scores referred to herein may differ from equivalent ratings developed and implemented by other ESG data providers in the market and may also differ from equivalent ratings developed and implemented by other divisions within the Deutsche Bank Group. Such ESG scores also differ from other ratings and rankings that have historically been applied in research reports published by Deutsche Bank AG. Further, such ESG scores do not represent a formal or official view of Deutsche Bank AG. It should be noted that the decision to incorporate ESG factors into any investment strategy may inhibit the ability to participate in certain investment opportunities that otherwise would be consistent with your investment objective and other principal investment strategies. The returns on a portfolio consisting primarily of sustainable investments may be lower or higher than portfolios where ESG factors, exclusions, or other sustainability issues are not considered, and the investment opportunities available to such portfolios may differ. Companies may not necessarily meet high performance standards on all aspects of ESG or sustainable investing issues; there is also no guarantee that any company will meet expectations in connection with corporate responsibility, sustainability, and/or impact performance.

Copyright © 2026 Deutsche Bank AG

This document is being provided for the exclusive use of YUI CHEUNG at THE UNIVERSITY OF HONG KONG.

David Folkerts-Landau
Group Chief Economist and Global Head of Research

Pam Finelli
Global Chief Operating Officer
Research

Steve Pollard
Global Head of Company
Research and Sales

Jim Reid
Global Head of
Macro and Thematic Research

Tim Rokossa
Head of Germany
Research

Gerry Gallagher
Head of European
Company Research

Matthew Barnard
Head of Americas
Company Research

Peter Milliken
Head of APAC
Company Research

Debbie Jones
Global Head of Sustainability
and Data Innovation, Research

Sameer Goel
Global Head of EM & APAC
Research

Francis Yared
Global Head of Rates Research

George Saravelos
Global Head of FX Research

Peter Hooper
Vice-Chair of Research

International Production Locations

Deutsche Bank AG
Deutsche Bank Place
Level 16
Corner of Hunter & Phillip
Streets
Sydney, NSW 2000
Australia
Tel: (61) 2 8258 1234

Deutsche Bank AG
Equity Research
Mainzer Landstrasse 11-17
60329 Frankfurt am Main
Germany
Tel: (49) 69 910 00

Deutsche Bank AG
Filiale Hongkong
International Commerce
Centre,
1 Austin Road West,Kowloon,
Hong Kong
Tel: (852) 2203 8888

Deutsche Securities Inc.
1-3-1 Azabudai
Azabudai Hills Mori JP Tower
Minato-ku, Tokyo 106-0041
Japan
Tel: (81) 3 6730 1000

Deutsche Bank AG
21 Moorfields
London EC2Y 9DB
United Kingdom
Tel: (44) 20 7545 8000

Deutsche Bank Securities Inc.
The Deutsche Bank Center
1 Columbus Circle
New York, NY 10019
Tel: (1) 212 250 2500

Deutsche Bank AG
Filiale Singapur
One Raffles Quay, South
Tower,
Singapore 048583
TeL: +65 6423 8001